cognizancejournal.com
Seun Adeoye, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.4, April 2025, pg. 77-104
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

Blockchain-Enabled, Post-Quantum Cryptographic Framework for Securing Electronic Health Records: A Next-Generation Approach to Healthcare Data Protection

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DOI: 10.47760/cognizance.2025.v05i04.006

Abstract: The fast digital transformation of healthcare systems has brought electronic health records (EHRs) into wide usage to enhance patient care and provide better data access. The need for better security grows more pungent as cybersecurity and quantum computing threats against traditional cryptographic approaches become more prevalent. This paper develops a Blockchain-Enabled Post-Quantum Cryptographic framework for protecting EHRs. The combination of blockchain technology with PQC safeguards healthcare data through decentralised distribution, unalterable data storage, and complete system transparency, and PQC prevents anticipated quantum computing vulnerabilities. Security and privacy improve in the proposed framework by combining lattice-based cryptography, hash-based signatures, and zero-knowledge proofs. Smart contracts enable the framework to enforce access policies and maintain regulatory compliance through its functionality. A performance analysis of this framework shows it can effectively secure EHRs through efficient and scalable implementation. The research demonstrates that PQC and blockchain offer healthcare organisations a secure protection solution for EHRs that fights evolving cyber threats within trustworthy healthcare systems.

Keywords: Blockchain, Post-Quantum Cryptography, Electronic Health Records, Healthcare Data Security, Smart Contracts, Zero-Knowledge Proofs, Lattice-Based Cryptography, Quantum-Resistant Encryption, Privacy-Preserving Healthcare Systems.

1. Introduction

1.1. Background of Electronic Health Records (EHRs)

Electronic Health Records (EHRs) have developed the healthcare industry through the digital handling of health-related data retrieval tools and patient information management capabilities. The healthcare system benefits from EHRs because they make medical history documentation, prescriptions, diagnosis reports, and treatment plans accessible to all medical facilities. This modern digital transformation system improves employee work efficiency while improving

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healthcare service delivery and lowering medical mistakes (Alabdulatif, 2025). Through interoperability, EHR systems permit healthcare providers to access patient data while performing analyses for evidence-based decision support and enhancing individual medical treatment plans.

EHR systems generate various advantages but create specific difficulties that primarily impact data protection and patient privacy scenarios. Patients' Medical data faces cyber threats due to the increased use of cloud-based and networked EHR platforms, which allow unauthorised access and expose sensitive data to breaches (Alabdulatif, 2025). Digital healthcare systems depend on patient record security because these records store sensitive information that demands absolute protection for doctors to maintain patient confidence.

1.2. The Need for Enhanced Security in EHR Systems

EHR systems have become vulnerable to security risks because healthcare data faces increased cyberattacks. Security breaches in EHR systems generate numerous negative results, including the theft of identities, insurance scheme scams, and privacy breaches for patients (Mahajan, 2022). The Health Insurance Portability and Accountability Act (HIPAA) in the U.S. and the General Data Protection Regulation (GDPR) in the European Union establish exact criteria that healthcare organisations must follow to protect their health information.

EHR systems have been successfully protected through RSA cryptography and ECC (Elliptic Curve Cryptography), which are conventional encryption methods. New quantum computer innovations create an immediate vulnerability for encryption methods that protect healthcare information systems (Miguel & Hannah, 2025). The powerful computational capability of quantum computers is a significant security threat to existing encryption methods, making currently available security techniques useless. This looming threat necessitates adopting quantum-resistant cryptographic solutions to future-proof healthcare data security.

Data stored in central locations exposes weak points, making it susceptible to hacking attempts and data manipulation due to single-entity failure issues (Sezer et al., 2025). Unauthorised medical record modifications result in medical malpractice that causes mistreatment and damages healthcare provider trust throughout the population. A better approach must be established to safeguard EHR through decentralised quantum-secure systems.

1.3. Role of Blockchain and Post-Quantum Cryptography (PQC) in Healthcare

Healthcare information security comes from using Blockchain combined with Post-Quantum Cryptography (PQC), which protects records against contemporary threats and upcoming cyber-attacks.

Blockchain Technology in EHR Security:

Blockchain succeeds as a decentralised immune ledger management system that provides secure solutions for EHRs. Combining cryptographic hashing with consensus mechanisms secures medical records by making them resistant to tampering and permitting controlled authorisation to access them. Using smart contracts brings enhanced security features through automatic access

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control enforcement that removes the necessity of middle parties while decreasing risks of data alterations.

Key benefits of blockchain in EHR security include (Sharma & Shambharkar, 2024):

- Decentralization: Standalone storage systems work together through decentralised methods that help prevent failure depending on central infrastructure.
- Immutability: The medical data storage system becomes indestructible because changes can only be made if all parties agree.
- Transparency and Auditability: The system features complete transparency, enabling healthcare stakeholders to detect all secure activities involving data access and modifications.
- Patient-Controlled Data Access: The technique enables people to protect their medical records through cryptographic keys so they can manage access permissions.

Post-Quantum Cryptography (PQC) for Future-Proof Security:

The security system PQC functions against quantum computing threats to ensure the safety of specific health information. The strength of PQC algorithms lies in their ability to use complex mathematical structures, such as lattice-based and hash-based cryptography, as well as multivariate polynomial cryptography defence against Shor's algorithm attacks (Thakur et al., 2024).

Blockchains integrated with PQC technology enable EHR systems to obtain three main security advantages (Sharma & Shambharkar, 2024):

- The usage of Quantum Resistance makes it possible to defend EHR data against quantum computing threats.
- Secure Key Management guarantees that authorised users have sole access to encrypted medical records and that their confidentiality is preserved.
- Zero-knowledge proofs (ZKPs), among other privacy-preserving computation techniques, allow secure data-sharing processes.

According to this research, a secure EHR protection framework is being developed by combining blockchain technology with PQC (Sharma & Shambharkar, 2024). This method will tackle existing and upcoming cyber threats to improve medical record security while maintaining confidentiality and accessibility in digital environments.

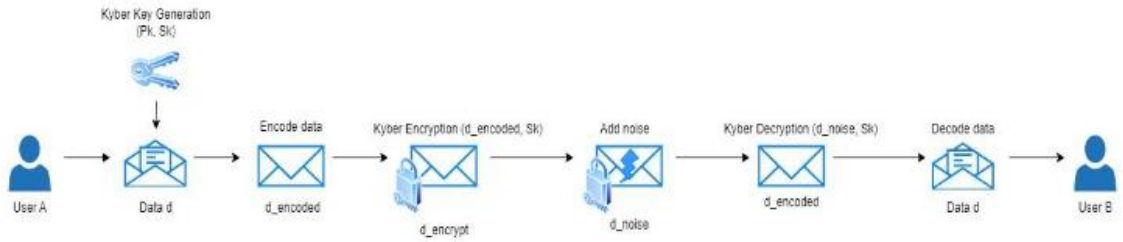

Figure 1: Data encryption using CRYSTALS Kyber

2. Current Security Challenges in Electronic Health Records

Modern healthcare depends on Electronic Health Records (EHRs) because they enable smooth data exchange between systems and improve patient results. Medical records digitisation introduced substantial security concerns that healthcare facilities must address (Sharma & Shambharkar, 2024). Modern cyberattacks against EHR systems have gotten more complex, while standard cryptographic methods prove inadequate to combat future threats because quantum computing advancements are now underway. The principal security problems affecting EHR systems are examined extensively in this section.

2.1. Cybersecurity Threats to EHRs

Medical records represent highly valuable targets in the healthcare sector because cyber attackers have made it their principal target. These represent the leading cybersecurity threats that affect the operation of EHRs (Anbarkhan, 2024):

2.1.1. Data Breaches and Unauthorized Access

Medical records filled with personally identifiable information (PII) and protected health information (PHI) make them desirable targets for cybercriminals to steal. Data breaches create three significant consequences: identity theft events, insurance fraud, and substantial monetary losses (Anbarkhan, 2024). Hacking occurs when criminals exploit insecure authentication systems or database misconfigurations with insider threats to take control of EHRs without authorisation.

2.1.2. Ransomware Attacks

Healthcare facilities experience extensive ransomware attacks because attackers encrypt their Electronic Health Records through malicious software and expect ransom payments for decryption. The attacks interfere with healthcare services, push back crucial medical treatments and deteriorate patient safety protocols (Anbarkhan, 2024). Hospital systems faced severe damage from the devastating WannaCry ransomware incident of 2017, among other high-profile attacks.

2.1.3. Man-in-the-Middle (MITM) Attacks

An MITM attack involves hackers who break healthcare provider-EHR systems communication channels to modify exchanged data. Malicious actors access patients' data while data manipulation and loss of confidentiality occur through such compromises (Anbarkhan, 2024). The exposure of medical data occurs through unsecured networks, weak encryption systems, and phishing attacks, enabling cybercriminals to execute MITM attacks.

2.1.4. Insider Threats and Human Errors

Advanced EHR security can be compromised through unintentional or intentional actions of healthcare staff members and employees within healthcare facilities. The internal threats affecting EHR security include the deliberate leaking of data, improper access attempts, and accidental mistakes (Anbarkhan, 2024). The lack of staff understanding regarding cybersecurity produces more EHR management weaknesses.

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2.1.5. Distributed Denial-of-Service (DDoS) Attacks

Unrestricted traffic patterns during DDoS attacks cause EHR systems performance degradation until medical data access is entirely blocked (Anbarkhan, 2024). Hospital operations become severely damaged when these attacks disrupt services, which extends the time needed for correct diagnosis and treatment.

The emergency exists for advanced security procedures to protect EHR data against the continuously evolving threat patterns.

2.2. Limitations of Traditional Cryptographic Methods

EHR security of EHRs uses two standard encryption methods: RSA (Rivest-Shamir-Adleman) and ECC (Elliptic Curve Cryptography). The current cryptographic methods encounter multiple necessary restrictions with their use.

2.2.1. Centralized Key Management Issues

Centralised encryption key servers in EHR systems attract cybercriminals because these servers serve as primary targets (Benaich et al., 2025). A successful compromise against the central key repository permits attackers to decrypt large amounts of EHR information while gaining the ability to modify it.

2.2.2. Vulnerability to Brute-Force Attacks

Traditional cryptographic algorithms become more vulnerable to brute-force attacks because advances in computing capabilities make it easier to perform such attacks (Benaich et al., 2025). Security gaps in key generation systems allow skilled attackers to break the encryption protection that protects patient data.

2.2.3. Lack of Data Integrity Mechanisms

The primary function of traditional encryption methods is protecting data confidentiality, although they provide insufficient methods to preserve data integrity (Benaich et al., 2025). Attackers infiltrate protected health records to modify their contents unbeknownst to detection personnel, who subsequently face inaccurate medical statements.

2.2.4. Scalability and Performance Bottlenecks

Healthcare data is proliferating, yet conventional encryption systems face challenges regarding performance expansion and calculation speed requirements (Benaich et al., 2025). Real-time processing of large EHR datasets affects system performance because of encryption-decryption operations that create bottlenecks.

Advanced security solutions must be developed to address cybersecurity threats against EHR systems and make them more resilient against attacks.

2.3. The Risk of Quantum Computing to Existing Security Protocols

Cryptographic systems face a complete breach potential because of the capabilities that quantum computing will provide. The speed of running mathematical computations by quantum computers depends on their qubits and quantum parallelism beyond traditional computer

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capabilities (Berwer, 2024). The technology makes it possible to break through cryptographic algorithms used by many systems today.

2.3.1. Breaking Traditional Encryption with Shor's Algorithm

The factorisation speed of quantum computers through Shor's algorithm enables them to break through RSA and ECC encryption methods (Berwer, 2024). RSA encryption that depends on integer factorisation difficulty and ECC encryption based on discrete logarithms will be vulnerable to polynomial-time attacks that quantum computers can perform.

2.3.2. Quantum Threat to Digital Signatures and Key Exchange Protocols

The current EHR systems verify users through cryptographic signature methods, which form the basis of their digital authentication systems (Berwer, 2024). Quantum computers can create fake digital signatures, resulting in unauthorised access to patient medical records.

EHR network security depends on Diffie-Hellman and RSA-based encryption tools as key exchange protocols (Berwer, 2024). These key cryptographic exchanges become vulnerable to quantum attacks, resulting in data eavesdropping and unauthorised interception.

2.3.3. Timeline for Quantum Threat Realization

Quantum computers remain under development, but serious threats appear likely to materialise in the next two decades based on recent advancements by Google, IBM and Chinese quantum scientists (Gandhi et al., 2024). Performing defensive measures to protect EHR systems from quantum security threats should be considered a pressing requirement rather than an issue for the future.

Electronic Health Records face growing security threats because of cyber-attacks, traditional encryption limitations, and the upcoming quantum computing risks that loom (Gandhi et al., 2024). The advancement of complex attack methods by attackers will cause conventional security models to become completely ineffective. Blockchain integration with post-quantum cryptography presents a practical solution that protects EHRs from the current and future threats that classical and quantum computing offer. The upcoming sections provide methods for these emerging technologies to build a resilient, future-proof security framework that protects healthcare data.

3. Blockchain Technology for Securing EHRs

An urgent need exists for better security systems in Electronic Health Record (EHR) systems to address escalating cybersecurity risks and system vulnerabilities. Blockchain technology is a promising security solution because it provides distributed accessible EHR data storage while remaining permanently unalterable across multiple nodes (Raiet al., 2024). Blockchain allows healthcare institutions to protect EHRs from unauthorised entry, data breaches and tampering. This section describes blockchain basics in healthcare systems, smart contracts, and decentralised access control practices and presents advantages in securing EHRs.

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3.1. Fundamentals of Blockchain in Healthcare

The distributive ledger technology (DLT) blockchain provides safe and transparent immutable record storage for decentralised networks. The system requires no centralised authority because it enables controlled access to secure EHR data storage. The healthcare blockchain technology implements these three fundamental principles (Raiet al., 2024):

3.1.1. Decentralization

Centralised EHR database configurations create vulnerability by allowing cyberattacks at weak single points of failure. Multiple node data distribution makes blockchain systems immune to unauthorised modifications as they guarantee continuous system availability.

3.1.2. Immutability

The network requires full agreement from all members to modify or remove data that gets written onto a blockchain. Data integrity in EHRs remains strong through this property because it blocks unauthorised modifications and stops fraudulent activities.

3.1.3. Cryptographic Security

The cryptographic hashing method and digital signatures form the core protection mechanisms of blockchain transactions (Raiet al., 2024). The cryptographic hash function running across blocks secures every entry, guarantees data authenticity, and resists tampering attempts on stored records.

3.1.4. Transparency and Auditability

The blockchain system provides complete visibility into EHR modifications, allowing stakeholders to examine recorded transactions for verification purposes (Raiet al., 2024). The healthcare systems benefit from increased regulatory compliance due to better accountability through blockchain applications.

3.1.5. Consensus Mechanisms

The consensus protocols Proof of Work (PoW), Proof of Stake (PoS), and Practical Byzantine Fault Tolerance (PBFT) lead blockchain networks to validate transactions (Raiet al., 2024). Only valid EHR modifications are recorded through these security procedures that protect the system.

Medical institutions utilise blockchain as a core infrastructure to provide protected healthcare data exchange and record access capabilities for patients, which helps to improve the deficiencies of standard EHR systems.

3.2. Smart Contracts and Decentralized Access Control

Blockchain-based technology permits self-executing digital agreements that run predefined rules without requiring external parties to facilitate them. Smart contracts support healthcare security by implementing automated access rules while maintaining integrity in the data-sharing policy.

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3.2.1. Role of Smart Contracts in EHR Security

Smart contracts enable secure connections between various healthcare parties consisting of (Peelam & Chamola, 2024):

- Patients regulate access to their medical records by deploying smart contracts that permit or deny permission to authorised individuals who wish to view their EHRs.
- Consent policies that healthcare providers must follow are enforced through smart contracts, which verify that providers meet HIPAA and GDPR requirements for accessing patient data.
- Smart contracts that use blockchain technology enable safe and secure communication of healthcare data between various medical institutions, including hospitals, clinics, and insurance providers.

3.2.2. Decentralized Access Control for EHRs

Centralised access control systems present two critical weaknesses because they depend on central authorities who face internal risks, and their systems have points at which failure can occur. The access control framework implemented by Blockchain uses decentralised methods that provide (Peelam & Chamola, 2024):

- The Role-Based Access method (RBAC) grants access through predefined roles, which include doctor, nurse, and insurance provider, and this system enforces access restrictions through smart contracts.
- The EHR data's detailed access control depends on user characteristics comprising job position, organisational scopes, and consent permissions through Attribute-Based Access (ABAC).
- Zero-knowledge proofs (ZKPs) running on blockchain enable entities to inspect information (like patient insurance qualifications) without disclosing their medical documents.

Smart contracts combined with the decentralised access control features of blockchain security systems guarantee complete protection for EHRs while upholding regulatory requirements.

3.3. Benefits of Blockchain for EHR Security

Blockchain integration in health care creates three key advantages: better security protocols, improved system accessibility, and enhanced patient decision rights. The following are key benefits (Peelam & Chamola, 2024):

3.3.1. Enhanced Data Security and Integrity

Medical records enjoy tamper-proof status since blockchain uses cryptographic security measures to protect them from unauthorised changes (Qadri et al., 2024).

The distributed system makes EHR databases less vulnerable to threats targeting centralised health record repositories.

3.3.2. Improved Patient Privacy and Control

- Every health record belongs to patients, who can establish complete control by selecting who accesses and shares their medical information.
- Patients can authorise safe identity verification protocols to stop the unauthorised sharing of medical records and protect their privacy.

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3.3.3. Resistance to Cyberattacks

- Blockchain protects against ransomware and man-in-the-middle (MITM) attacks throughout the process, removing all single failure points within the system.
- Strict security rules within smart contracts minimise data breaches resulting from internal threat actors and accidental human mistakes.

3.3.4. Secure Interoperability and Data Sharing

- Through its system, hospitals, laboratories and insurance providers conduct safe data exchanges without intermediary involvement.
- The distributed ledger accepts input contributions only from healthcare entities with verification status through consensus processes.

3.3.5. Compliance with Regulatory Standards

- Blockchain EHR implementations follow HIPAA, GDPR, and additional regulatory standards, ensuring secure data processing and medical information storage.
- Immutable audit trails allow regulatory bodies to track and confirm healthcare compliance through system operations.

3.3.6. Fraud Prevention and Insurance Transparency

- Through blockchain verification of claims against unalterable medical records, insurance companies minimise fraud when releasing payments.
- Healthcare transactions can avoid medical billing fraud and identity theft because blockchain provides identity verification services.

The security solution of Electronic Health Records requires Blockchain technology because it distributes data through decentralisation while ensuring data immutability with cryptographic security and transparency features. Using smart contracts and decentralised access control systems strengthens EHR privacy by providing protected data exchange capabilities, automatic consent processes, and user access permissions control (Qadri et al., 2024). As a result, blockchain presents an effective structural solution to combat the security problems that conventional EHR systems encounter.

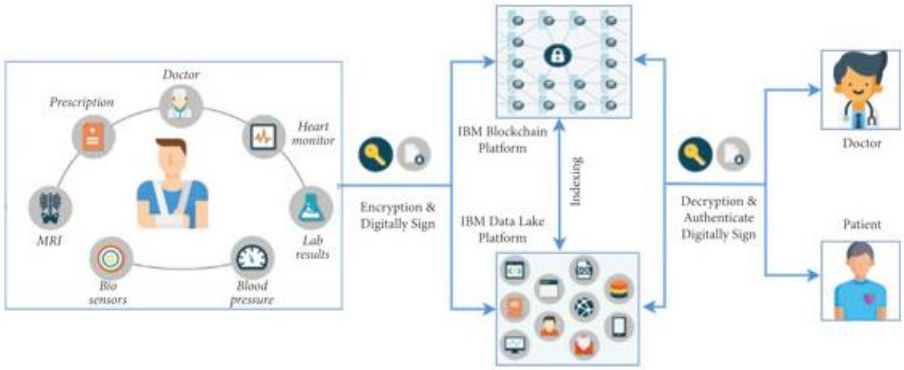

Figure 2: A Blockchain Framework to Secure Personal Health Record (PHR)

4. Post-Quantum Cryptographic Framework

The emergence of quantum computing threatens widely used encryption approaches, which require healthcare institutions to implement quantum-resistant security solutions for their Electronic Health Records. PQC is a breakthrough cryptographic approach to creating algorithms that can withstand quantum computer-powered attacks (Qadri et al., 2024). When implemented with blockchain technology, PQC achieves superior health data protection by establishing complete protection against present-day and upcoming cyberattacks.

4.1. The Need for Post-Quantum Cryptography in Healthcare

The documented computational methods inside quantum computers use superposition alongside entanglement principles to easily outperform classical computers in solving mathematical problems at quantum speeds (Robert et al., 2024). The computational power behind quantum technology weakens other encryption techniques, including RSA and ECC, which are susceptible to breaking attack methods. RSA encryption falls victim to Shor's algorithm because it efficiently factors large prime numbers, and Grover's algorithm makes brute-force attacks more efficient, thus weakening symmetric encryption schemes.

Healthcare institutions depend on quantum-resistant security because they manage the constant exchange and storage of sensitive patient data. EHR system breaches create multiple security risks, including unauthorised access to records, patient data theft, and alterations of vital medical information (Robert et al., 2024). Healthcare institutions implementing PQC within their security frameworks will minimise these risks while protecting data during long periods.

4.2. Key Post-Quantum Cryptographic Algorithms for EHR Security

A collection of PQC algorithms exists as suitable alternatives to traditional cryptographic measures. Such algorithms sustain their strength against classical and quantum attack methods using advanced mathematical constructions.

4.2.1. Lattice-Based Cryptography

Lattice-based cryptography represents a top PQC candidate because it offers excellent security performance and operational efficiency. Mathematical problems involving shortest vectors in high-dimensional lattices serve as the foundation for this system because quantum computers struggle to resolve them (Yi, 2023). The method enables the secure transmission of keys, encryption algorithms, and digital signature capabilities that suit EHR security functions.

4.2.2. Hash-Based Cryptography

The trusted security of cryptographic hash functions allows Hash-based cryptography to produce strong digital signatures (Yi, 2023). The technology finds strong use in blockchain operations because blockchain requires both immutable records and verifiable data. XMSS (eXtended Merkle Signature Scheme) belongs to the hash-based cryptography category that fights quantum attacks and maintains operational efficiency.

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4.2.3. Code-Based Cryptography

The cryptographic technique, code-based cryptography, provides secure encryption and authentication by implementing error-correcting codes. Atomic Key Exchanges use keys with substantial sizes to provide robust protection that suits medical information safety requirements (Yi, 2023). An example encryption scheme from the code-based cryptography category is known as Classic McEliece.

4.2.4. Isogeny-Based Cryptography

Post-quantum key exchange mechanisms find a strong candidate in isogeny-based cryptography because the security depends on the difficulty of finding isogenies between supersingular elliptic curves. The scheme creates fast and protected channels to establish encrypted communication networks for healthcare environments (Singh et al., 2024). Educational institutions using these PQC algorithms in their EHR systems receive quantum-resistant security protection, which protects patient data from modern cyberattacks.

4.3. Integrating Post-Quantum Cryptography with Blockchain for EHR Protection

When PQC technology joins blockchain, it develops a thorough security system that guards EHRs against unauthorised disclosure while preserving their completeness and accessibility (Singh et al., 2024). Blockchain's decentralised ledger solution for secure data storage gains additional quantum security through PQC implementation.

4.3.1. Enhancing Blockchain Security with Quantum-Resistant Cryptographic Techniques

Standard blockchain networks protect transactions and other core cryptographic operations through signature technologies, consensus formation methods, and cryptographic methods. Several current cryptographic network protocols exist that quantum attacks could break (Singh et al., 2024). Blockchains maintain secure digital signature functions through the deployment of Dilithium and XMSS protocols since they represent quantum-resistant alternatives to RSA and ECC encryption.

4.3.2. Quantum-Safe Smart Contracts for Healthcare Data Management

The critical function of smart contracts in blockchain-enabled EHR systems consists of automating data access protocols and sharing functions. Smart contracts gain improved security through PQC because they implement quantum-resistant authentication schemes. ZKPs, when integrated with PQC, create access control systems that allow patients to demonstrate their medical service eligibility while maintaining their data privacy.

4.3.3. Secure Key Exchange and Hybrid Encryption for EHR Protection

Combining PQC algorithms with hybrid encryption offers exceptional security and performance compromise for encryption methods. The solution combines PQC key exchange mechanisms to create secure communication lines through which symmetric encryption technology, such as AES-256, encrypts large medical data records (Singh et al., 2024). The healthcare system can use Blockchain technology to distribute encryption keys securely so authorized personnel gain access to patient medical documentation.

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4.4. The Future of PQC-Enabled Blockchain Security for EHRs

Medical institutions must join efforts to develop quantum-resistant security frameworks now because the growth of quantum computing threatens current security standards. The healthcare field must actively introduce PQC-integrated blockchain systems to protect EHRs from developing security threats (Agarwal et al., 2024). Government entities, research establishments, and cybersecurity specialists need to work together to implement standardized PQC procedures and create efficient cryptographic standards for healthcare settings.

Modern EHR systems secured with PQC-enhanced blockchain tools gain sustained protection from classical and quantum-based cyber threats. Protecting patient data privacy and integrity requires the combination of quantum-resistant digital signatures and both post-quantum key exchanges and unalterable blockchain records (Agarwal et al., 2024). The following section outlines an implemented blockchain-based post-quantum cryptographic framework for protecting EHRs, including its design, deployment methods and assessment results.

5. Design and Architecture of the Proposed Framework

To integrate the blockchain and PQC-based security with Electronic Health Record (EHR) systems in a secure manner, an appropriate framework is required that protects the confidentiality of the healthcare data while preserving integrity and providing access control against the quantum attacks (Agarwal et al., 2024). This suggested framework attempts to implement blockchain decentralization using PQC algorithms, which likewise enhances the mechanisms of authentication and access control and secures the data. In this part, we describe the model design of the system and how its encryption features and consensus-based protocols for processing of safe transactions under EHR system are used towards doing the same.

5.1. System Model and Components

These components are considered for creating a suggested post-quantum cryptographic framework built upon blockchain technology and used to enhance the security levels of HER (Agarwal et al., 2024). The patients, healthcare providers, and insurers can interact on a secure decentralized blockchain ecosystem, allowing them to manage the system transactions.

5.1.1. Blockchain Network for EHR Management

By the foundation of blockchain, there is no alteration in patient records and transaction logs, and cryptographic keys are kept securely in a distributed ledger system. To protect the data from unauthorized modifications, each block contains encrypted patient references, timestamp information, and cryptographic validation statements (Agarwal et al., 2024). In permissioned access blockchains, access is through limited permission to authorized entities only so that only authorized parties are the network participants.

5.1.2. Post-Quantum Cryptography Layer

The PQC layer is a replacement of the encryption operating within the system to design the protective schemes resistant to quantum attacks. The authentication and the data transmission are

protected by using lattice-based and hash-based cryptography for the key m-ouncement system based on PQC algorithms (Agarwal et al., 2024). The PQC layer is used to keep a secure system of encryption keys so that they stay protected against quantum decryption attacks.

5.1.3. Smart Contracts for Access Control

The smart contracts allow the implementation of data-sharing procedures and access controls using automatic procedures inside the system. These are policies that define systems around permissions based on role (which can be any role) and which sections of medical records patient consent get access to (Agarwal et al., 2024). As the transaction authentication method, the system uses quantum-safe digital signatures that verify validity under an attack of quantum computing.

5.1.4. Secure Data Storage and Retrieval Mechanism

EHRs are stored via back-off-chain storage, i.e., using secure cloud solutions or the InterPlanetary File System (IPFS). In addition, the blockchain system holds cryptographic hashes and access credentials along with the encrypted off-chain EHR storage (Agarwal et al., 2024). This methodology provides system growth, cost-effective data storage, and data reliability across the whole system.

5.2. Encryption and Data Access Mechanisms

The security and confidentiality of EHRs are secured using post-quantum cryptography. In this respect, EHRs are stored on the blockchain utilizing access control systems using Blockchain.

5.2.1. Post-Quantum Encryption for EHR Security

Credit Suisse implements quantum-resistant encryption systems to encrypt stored and transferred medical records for protection. Lattice-based cryptographic methods are employed to provide secure authentication, while the key exchange and digital signature functions use Lattice-based cryptography simultaneously (Wenhua et al., 2023). AES-256 encryption is used as a symmetric method for protecting data size and PQC to secure encryption keys in medical data.

5.2.2. Role-Based and Attribute-Based Access Control

Data accessibility in the framework is managed by the dual role-based and attribute-based access control (RBAC-ABAC) method. Besides, the system uses predefined roles (i.e. doctors, nurses and administrators) for granting access permissions and contextual elements, including emergency conditions, location and zeitgeist, to determine access permissions (Wenhua et al., 2023). To ensure the security of patient data, it dynamically configured the access control policies to give admission to the authorized users.

5.2.3. Secure Key Management and Authentication

The key management system, which binds blockchain technology with PQC, makes it impossible for unauthorized access because it is a decentralized approach. Threshold cryptography enables blockchain nodes to have a slice of the encryption keys to prevent a centralized storage model (Wenhua et al., 2023). With the agreements done through smart contracts, the healthcare providers can access the data briefly while the patient retains control of the encryption key.

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5.2.4. Data Integrity and Auditability

Electronic Health Records can be changed, or tried to access them at which the blockchain entries remain unalterable. An unalterable trail that may be used by patients and regulators to verify that the records are secure is included (Wenhua et al., 2023). Digital signatures based on hash produce alerts whenever any data of storage is tampered with.

5.3. Consensus Protocol for Secure Data Exchange

The system employs an agreement procedure with fast secure message exchange among parties that does not compromise security, confidentiality, or data reliability. Proof of Work (PoW) is very resource intensive, which is not very suitable for healthcare data protection since the data is sensitive (Oliva et al., 2024). Rather than using standard consensus options, a specific hybrid consensus model was created.

5.3.1. Practical Byzantine Fault Tolerance (PBFT) for Secure Transactions

It uses the PBFT algorithm as the main consensus mechanism of the framework. PBFT was helpful to the permissioned blockchain network because it ensured fast operational speed and minimal delay times (Oliva et al., 2024). The predefined mechanism of validation validates every transaction concerning EHR access and modifications by a set of trusted nodes to keep the system decentralized, protecting it from malicious threats.

5.3.2. Quantum-Resistant Digital Signatures for Authentication

The consensus protocol relies on the CRYSTALS—Dilithium quantum resistant algorithm, among others, which allows the consensus protocol to authenticate transactions using a digital signature method (Oliva et al., 2024). Quantum-resistant standards are used with the blockchain network's electronic signatures to combat system fraud and mark every transaction performed on the network.

5.3.3. Secure Multi-Party Computation (SMPC) for Privacy-Preserving Data Sharing

Within the framework, patient data are protected from privacy during their safe exchange using Secure Party Computation (SMPC). SMPC allows multiple entities to perform a joint computation on their distinct inputs while no single data is exposed to others (Oliva et al., 2024). This system allows healthcare providers who are supported by it to exchange analytical findings from electronic health records while protecting both, privacy and the research capabilities at the same time.

5.3.4. Zero-Knowledge Proofs (ZKPs) for Enhanced Privacy

Privacy-safe authentication techniques, i.e., zero-knowledge Proofs, are utilized to support authentication in the entire consensus protocol framework. ZKPs are used to provide user credential access to the verifier without the verifier ever learning about any additional bits of data (Oliva et al., 2024). The strict privacy rules for health care prevent it from being exposed or used by unauthorized means, and the method cuts down on this potential.

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5.4. Summary of the Proposed Framework

This proposed post-quantum cryptographic framework, using blockchain technology, enhances EHR security and privacy and also ensures integrity via quantum-resistant healthcare systems that will operate upon the coming of age of quantum computing (Arastouei & Khan, 2025). As proposed in the framework, this new advanced security solution combines PQC quantum-proof encryption methods with blockchain decentralization and data tamper-proof features.

This model consists of various core elements, which are as follows (Arastouei & Khan, 2025).

- A network of licensed blockchains for use in high security for medical record treatment is created.
- The data is protected from quantum attacks through the use of a post-quantum procedure encryption method.
- Smart contracts for role-based and attribute-based access control.
- Encryption keys are managed by a decentralized system that preserves safe user verification.
- PBFT-based consensus with PQC digital signatures for secure transaction validation.

Moreover, the system is equipped with Zero (ZK) Proof and Secure Multi Party Computation (SMPC) to keep the data sharing private.

The proposed framework has two purposes: it defends healthcare organizations against current security threats and quantifies resistant protection measures being made to protect against future security threats. The next part of the explanation regards the deployment strategies for healthcare operations and the framework implementation and performance testing process.

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Seun Adeoye, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.4, April 2025, pg. 77-104
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

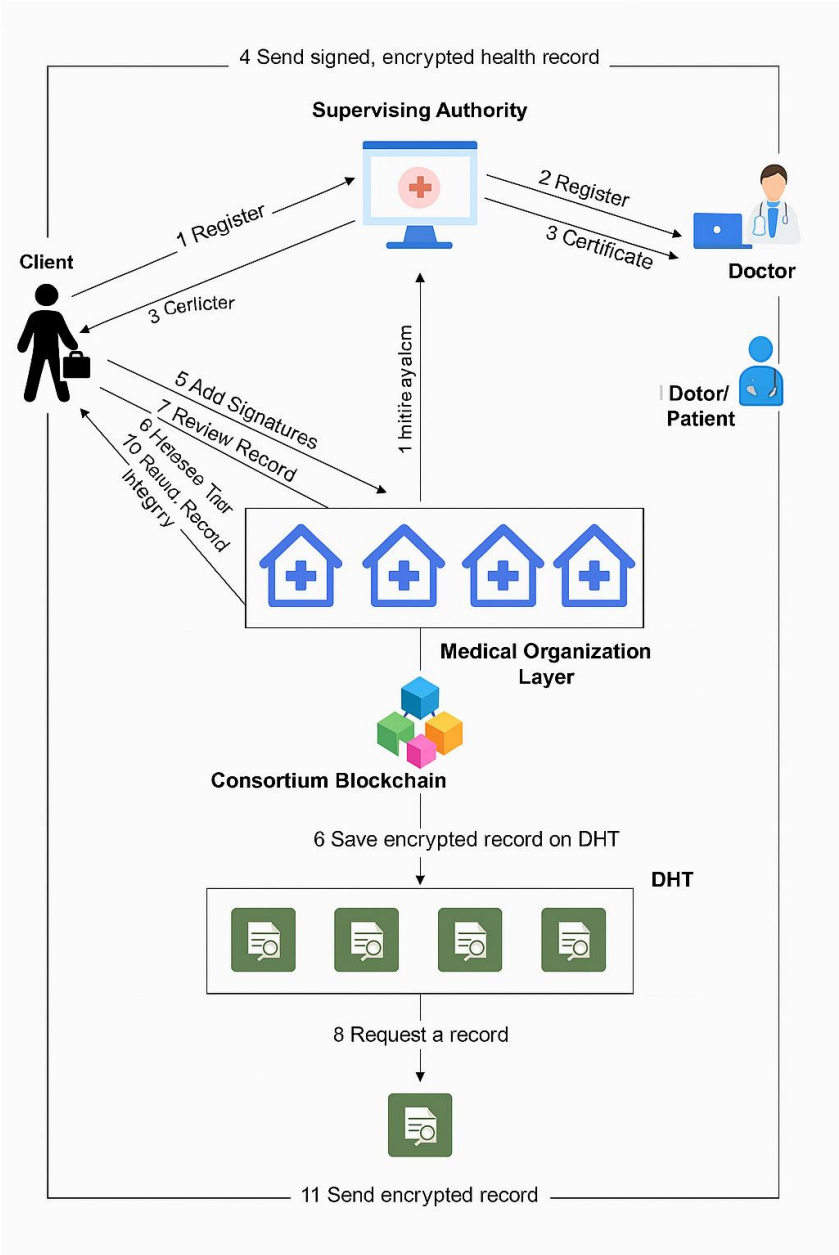

Figure 3: System representation for the proposed PQB based EHR Framework

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6. Implementation and Performance Evaluation

A rigorous analysis needs to evaluate the effectiveness, security, and efficiency of the proposed PQC-based blockchain framework when handling EHRs (Arastouei & Khan, 2025). The following section outlines simulation and testing methods, security and efficiency analysis of the system, and a similar evaluation of classic security models, demonstrating the proposed framework's benefits.

6.1. Simulation and Testing Approaches

The implementation team develops a simulated environment based on blockchain platforms and cryptographic libraries that support the execution of post-quantum cryptographic algorithms (Arastouei & Khan, 2025). The implementation process requires permissioned blockchain network setup and integration of PQC techniques while developing smart contracts for access control.

6.1.1. Blockchain Network Deployment

Hyperledger Fabric is the platform for implementing the blockchain network because it delivers permissioned blockchain functionality needed for transactional security and scalability (Arastouei & Khan, 2025). The network includes various peer nodes which symbolize healthcare providers, patients, and insurers. The ledger maintains a permanent record of every EHR access event, update changes, and shared data.

6.1.2. Integration of Post-Quantum Cryptographic Algorithms

The framework implements PQC algorithms as an essential component for providing quantum-resistant security through the selection of these systems (Arastouei & Khan, 2025):

- The digital signature authentication mechanism depends on the CRYSTALS-Dilithium alongside Falcon cryptographic algorithms.
- The framework uses Kyber as its key encapsulation system to guarantee secure device connection.
- A blockchain hashing system and integrity verification capabilities depend on XMSS (eXtended Merkle Signature Scheme).
- A combination of algorithms from Open Quantum Secure libraries tests post-quantum cryptographic primitives for practical implementations.

6.1.3. Smart Contract Development and Access Control

The development process for smart contracts uses Solidity to create programs deployed within Hyperledger Fabric chaincode. The defined access control policies use roles to determine which authorized entities can access particular medical records through the contracts. Testing includes (Bhattacharya & Bansal, 2025):

- An examination of smart contracts' execution speed takes place during the testing procedure.
- The system must automatically change access permissions according to healthcare policies that have been established in advance.

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- The assessment process covers the ability to work together with current electronic health record systems.

6.1.4. Simulation of Quantum Attacks

The security strength of the framework undergoes a simulation-based analysis between classical and quantum computing systems. Quantum algorithms Shor's and Grover's facilitate the testing of traditional cryptography's susceptibility to attacks after PQC encryption implementation (Bhattacharya & Bansal, 2025). The framework testing method through quantum decryption attack validation ensures assessment of its ability to resist decryption breakthroughs in the quantum era.

6.2. Security and Efficiency Analysis

The security evaluation measures framework performance through essential metrics, including data protection measures, transaction duration, cost, and attack defence capabilities.

6.2.1. Security Assessment

A security evaluation of the framework occurs by subjecting it to diverse attack vectors through testing which includes (Bhattacharya & Bansal, 2025):

- The blockchain network protects Electronic Health Records (EHR) from man-in-the-middle attacks by being immutable and by using PQC-secured encryption keys.
- Data tampering and breaches become impossible to execute because the combination of hash-based digital signatures and blockchain storage prevents any unauthorized manipulation of records.
- The framework still maintains strong encryption strength against quantum computing attacks since it uses lattice-based and hash-based cryptography, which make quantum codebreaking operations impractical.

6.2.2. Efficiency Metrics

The proposed framework shows its operational performance through essential Key Performance Indicators that include these assessments (Atal et al., 2024):

- Transaction throughput demonstrates EHR's processing ability since it measures the speed at which medical records interact with the system to show how well it scales.
- A system performance analysis studies PQC algorithm computational requirements to understand their impact on system operations.
- Smart contract execution times are evaluated through measuring both the validation duration for access permissions and the retrieval time of encrypted records.
- The implementation of PQC-based encryption increases processing load but hybrid encryption techniques with symmetric key methods can decrease its impact on performance speed.

6.3. Comparative Study with Traditional Security Models

Evaluation of the proposed framework benefits starts with a comparison between blockchain-enabled PQC implementation and standard EHR security systems built with traditional cryptographic approaches (Atal et al., 2024).

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Table 1: Comparative Study with Traditional Security Models

Security Feature	Traditional Security Models	Blockchain-Enabled PQC Framework
Encryption Mechanism	RSA, ECC, AES	Post-Quantum Cryptography (Lattice-based, Hash-based, Code-based)
Quantum Resistance	Vulnerable to quantum attacks	Resistant to quantum decryption algorithms
Data Integrity	Centralized storage prone to tampering	Blockchain immutability ensures tamper-proof records
Access Control	Role-based access with centralized servers	Smart contracts enforcing decentralized access control
Transaction Transparency	Limited auditability due to centralization	Fully auditable and transparent record transactions
Scalability	Limited by centralized architecture	Distributed ledger ensures higher scalability
Attack Resistance	Susceptible to MITM, ransomware, and database breaches	Resistant to quantum attacks and data tampering

Key Findings from the Comparative Study

1. PQC algorithms create an additional layer of security which maintains EHR data safety even when quantum computers gain practical use capabilities.
2. The blockchain-based method provides decentralized access control because it removes the requirement for central authentication authorities.
3. Since patient records exist on an unalterable ledger the system provides full verification capability for all record access attempts as well as modifications.
4. When implemented through hybrid encryption PQC maintains performance standards that optimize security effectiveness for practical healthcare use.

6.4. Summary of Implementation and Performance Evaluation

This proposed framework reveals its capabilities to become an adaptable security design for EHRs. Quantum-resistant cryptographic techniques demonstrate successful integration with blockchain through simulations, enabling decentralised access control and data confidentiality and integrity (Atal et al., 2024). Performance evaluation shows PQC encryption produces greater computational overhead than commonly used cryptographic systems but hybrid encryption and efficient blockchain consensus protocols help minimize performance issues.

These are the main evaluation results that have been identified (Atal et al., 2024):

- The security framework operates as quantum-resistant through the combination of PQC algorithms which include lattice-based and hash-based and code-based methods.

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- Using blockchain technology provides clients with impermeable medical records that feature full transparency through audit checks.
- Through smart contracts the framework implements rigorous authentication procedures that limit unauthorized entry points.
- This model offers better resistance against traditional and quantum-based cyber threats when considering conventional security approaches.

The second section includes an analysis of conclusion and future development prospects regarding blockchain-enabled post-quantum cryptographic security for EHR systems.

7. Ethical, Legal, and Regulatory Considerations

A blockchain-enabled PQC framework for securing EHRs creates multiple ethical and regulatory challenges in addition to legal concerns. Medical health data demands strict obedience towards worldwide privacy laws coupled with ethical patient rights management while addressing actual implementation challenges (Adeyinka & Adeyinka, 2025). This section investigates regulatory environments, ethical factors, and implementation barriers that emerge from using this emerging technology in healthcare systems.

7.1. Compliance with Healthcare Data Privacy Laws

Every EHR security framework aimed at preserving patient information confidentiality must meet existing healthcare privacy standards for both security and confidentiality (Adeyinka & Adeyinka, 2025). The blockchain-PQC framework meets critical data protection requirements that govern the United States healthcare sector through the Health Insurance Portability and Accountability Act (HIPAA) while satisfying the General Data Protection Regulation (GDPR) protocols of the European Union.

7.1.1. HIPAA Compliance

HIPAA mandates strict security and privacy requirements for handling Protected Health Information (PHI). The proposed framework meets demand through its following features (Adeyinka & Adeyinka, 2025):

- The digital records' immutability function of Blockchain helps maintain patient data integrity thus fulfilling HIPAA's obligation for keeping audit trails.
- Smart contracts implemented through the proposed framework provide role-based access controls which authorize healthcare professionals to either view or edit EHRs.
- The implementation of PQC-based encryption generates superior PHI confidentiality results than what HIPAA requires for secure data transmission and storage.

7.1.2. GDPR Compliance

The data protection concepts of GDPR align with requirements to obtain patient consent while providing the right to forget data processing records. The proposed framework adheres to GDPR principles through (Mahajan & Reddy, 2024):

- The system allows patients to govern healthcare record access through cryptographic keys which satisfies GDPR user consent standards for personal data control.
- Security challenges related to the right to be forgotten are addressed by blockchain immutability except for off-chain storage where Zero-Knowledge Proofs (ZKPs) alongside cryptographic methods help users selectively erase their data while maintaining system security.
- Blockchain's distributed framework allows safe data sharing between medical institutions through a decentralized architecture that satisfies GDPR data portability obligations.

7.1.3. Other Regional Regulations

Several global rules including the Personal Data Protection Act (PDPA) in Singapore along with the China's Personal Information Protection Law (PIPL) and the Kenya's Data Protection Act (DPA) establish mandatory guidelines for health data management.

7.2. Ethical Implications of Decentralized Health Data Management

Blockchain technology alongside PQC creates secure systems however evidence points to ownership disputes about medical data as well as individual patient rights and healthcare technology distribution fairness problems (Gharaviet al., 2024).

7.2.1. Patient Autonomy and Data Ownership

Through decentralization medical patients gain authorship over their records so they no longer need to depend on central medical organizations to store or retrieve healthcare data. Patients face ethical difficulties in using encryption keys due to their lack of technical understanding which can result in healthcare data becoming unrecoverable (Mahajan & Reddy, 2024). The framework resolves this issue through robust access management capabilities combined with protected recovery procedures that protect autonomy without sacrificing ease of use.

7.2.2. Data Privacy vs. Transparency

The transparent characteristics of blockchain systems diminish patient privacy requirements in healthcare fields. Healthcare transaction data transparency to network participants even through hashed record viewing creates privacy protection risks for individual patient records (Mahajan & Reddy, 2024). The framework tackles this issue through ZKPs and SMPC protocols that let users verify transactions while keeping confidential data protected from disclosure.

7.2.3. Risk of Data Misuse

Decentralized health data storage systems which implement robust encryption still face ethical vulnerabilities when unauthorized users manage to access protected information. Insurers and employers along with third parties possess the capability to use patient records against patients in discriminatory ways (Mahajan & Reddy, 2024). Healthcare ethical practices rely on fine-grained access control policies as well as smart contract-enforced consent management to maintain ethical data access.

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7.3. Challenges in Adoption and Practical Implementation

Healthcare organizations must overcome multiple implementation barriers when integrating blockchain along with PQC into their current systems because these cryptographic solutions provide valuable security features.

7.3.1. Technical Integration with Legacy Systems

Healthcare facilities depend on centralized electronic health records systems which were created prior to decentralized systems. Integrating blockchain demands extensive modifications to present-day infrastructure bases which involve three primary aspects (Akoramurthy & Surendiran, 2024):

- The implementation of interoperable application programming interfaces serves as a solution to link traditional database systems with blockchain networks.
- Backward compatibility should be maintained throughout the process to stop healthcare service delivery from being interrupted.
- The effectiveness of blockchain-enabled EHR systems requires IT personnel and healthcare professionals to receive proper training for management responsibilities.

7.3.2. Computational and Storage Overhead

The cryptographic algorithms implemented after quantum computing security methods demand more computational resources compared to standard encryption standards. Hash-based cryptography and lattice-based cryptographic systems need additional processing resources to perform authentication and retrieve data thus causing delays in the process. The storage capacity required to operate blockchain transactions becomes substantial because the system maintains several duplicate records of the ledger. Possible solutions include (Akoramurthy & Surendiran, 2024):

- System deployment utilizes PQC algorithms to protect key exchange but adapts AES-256 symmetric encryption for secure large dataset protection.
- Blockchain storage volume decreases through the deployment of off-chain storage systems consisting of IPFS technology and cloud storage solutions.

7.3.3. Regulatory and Legal Uncertainty

Healthcare regulations remain vague about blockchain protocols along with PQC technology which produces uncertain conditions regarding compliance standards and legal responsibilities. The current policies must receive updates from policymakers to support decentralized healthcare models without creating barriers to innovation (Akoramurthy & Surendiran, 2024). To build clear legal frameworks the three groups of blockchain developers and healthcare organizations and regulatory bodies must strengthen their collaborative efforts.

7.3.4. Adoption Resistance and Stakeholder Concerns

Healthcare institutions exhibit a slow pace in implementing new technologies mainly because of worries regarding safety measures and financial expenses and regulatory requirements. Resistance may arise from (Akoramurthy & Surendiran, 2024):

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- Then come healthcare providers who show caution about changing their existing frameworks for blockchain-based systems.
- Healthcare patients tend to resist self-managing their medical history by using cryptographical key management methods.
- Insurers together with policy-making authorities possess concerns regarding blockchain-based EHRs due to their regulatory implications.
- The adoption of blockchain-PQC security needs programs with real-life implementations and awareness programs and financial benefits to demonstrate advantages in healthcare settings.

7.4. Summary of Ethical, Legal, and Regulatory Considerations

The deployment of this proposed EHR management framework requires resolving various ethical, legal and adoption obstacles to deliver highly secure and future-proof EHR solutions.

Key takeaways include (Akoramurthy & Surendiran, 2024):

- The framework maintains compatibility with global healthcare data privacy laws such as HIPAA, GDPR, and regional regulations to protect patient records during legitimate access.
- The decentralised system enhances patient self-governance yet creates new privacy concerns which need to be managed by implementing cryptographic defence mechanisms consent-based resource sharing and precise access control systems.
- Healthcare adoption of blockchain technology with PQC demands essential updates of systems and regulatory standards and solutions for stakeholder opposition to succeed.

The success of blockchain-enabled post-quantum security deployments in healthcare needs proper attention to these important issues. A concluding analysis of this paper follows to summarise important findings while proposing future research paths for EHR security improvement in quantum computing environments.

8. Future Directions and Conclusion

Modern healthcare data management has experienced an essential change by combining blockchain technology with post-quantum cryptography (PQC) to secure Electronic Health Records (EHRs). Despite its ongoing evolution this field creates new problems while also producing beneficial options for future development (Natarajan et al., 2025). The final section will discuss healthcare blockchain innovations as well as PQC advances while reviewing current security challenges and opportunities for EHRs before delivering research conclusions.

8.1. Potential Advancements in Blockchain and PQC for Healthcare

The secure handling of EHRs through Blockchain and PQC technology will grow better because of ongoing development.

8.1.1. Scalability Enhancements in Blockchain for Healthcare

The major barrier encountered when implementing blockchain is its inability to scale properly. Future advancements may include (Natarajan et al., 2025):

- The sharding method splits blockchain networks into distinct sections, enabling multiple transaction processing operations to enhance total network speed.
- bordelayer-2 scaling solutions consisting of state channels and sidechains help transfer transactions off the main blockchain to improve its efficiency.
- DAG-based blockchain models provide an adjustable network structure that speeds up transaction confirmations while boosting network throughput.

8.1.2. Quantum-Secure Blockchain Protocols

- Blockchains need to implement PQC-resistant protection mechanisms in their protocols because quantum computing continues to grow stronger.
- Proof-of-Work (PoW) and Proof-of-Stake (PoS) are replaced by quantum-resistant consensus algorithms in order to protect blockchain security.
- Post-Quantum Digital Signatures consisting of hash-based and lattice-based cryptographic signatures should replace the ECC and RSA signature methods.
- The technology allows secure computation of encrypted EHRs through homomorphic encryption so operations run without exposing patient data.

8.1.3. AI-Driven Security for EHRs

Blockchain and PQC systems receive performance improvement through artificial intelligence (AI) and machine learning (ML) solutions which boost threat monitoring systems and access control capabilities. AI-driven solutions can (Natarajan et al., 2025):

- The system uses real-time analysis of blockchain transaction patterns for detecting and responding to cybersecurity threats.
- Security audits of smart contracts should be automated to find vulnerabilities before contractors release their contracts into operation.
- The implementation should optimize PQC encryption procedure for efficient performance without compromising security.

8.1.4. Interoperability and Standardization

The future of research should concentrate on the following aspects to simplify blockchain-PQC integration adoption.

Future research should target the development of worldwide standards which enable healthcare systems to safely exchange information when integrating blockchain-PQC technologies.

Interoperability protocols between blockchain networks will securely enable different blockchain platforms such as Hyperledger and Ethereum and private blockchains to share health records.

A regulatory-compliant framework exists to check that decentralized EHR systems follow evolving privacy laws and compliance requirements.

8.2. Challenges and Opportunities in Securing EHRs

The successful implementation of both blockchain and PQC cryptography for EHR security needs multiple fundamental obstacles to be solved (Natarajan et al., 2025).

8.2.1. Challenges in Securing EHRs

1. **Computational and Storage Overhead:**
   - The implementation of PQC algorithms requires extensive computational resources which leads to decreased healthcare system operational speed.
   - Due to its need for multiple ledger copies the blockchain system creates storage challenges which affect scalability abilities.
   - Research should investigate mixed encryption methods and improved consensus algorithms which reduce operational costs in future blockchain applications.
2. **Regulatory and Compliance Hurdles:**
   - The unalterable nature of blockchain systems creates difficulties for meeting GDPR requirements concerning "Right to Be Forgotten" while new solutions involving off-chain storage and ZKPs help overcome this.
   - A shortage of well-defined regulatory guidelines about blockchain in healthcare systems results in uncertainties which require collaborative actions between healthcare specialists and technology developers alongside policymakers.
3. **Resistance to Adoption and Technological Barriers:**
   - Healthcare institutions avoid shifting from centralized to decentralized EHR systems because they fear the implementation costs along with technical difficulty and integration problems.
   - More healthcare personnel would start adopting new systems through performance-based incentives combined with targeted educational initiatives.

8.2.2. Opportunities for Innovation and Growth

1. **Enhanced Patient Empowerment and Data Ownership:**
   - The decentralized approach of blockchain technology allows patients to control their medical records which results in greater transparency along with healthcare system trust.
   - The access control mechanism based on smart contracts enables authorized data sharing when patients explicitly consent to the process which respects ethical guidelines in data governance.
2. **Collaboration Between Academia, Industry, and Governments:**
   - The implementation of blockchain-PQC integration in real-world medical environments requires research institutions, healthcare organizations and regulatory bodies to form partnerships that conduct pilot projects.
   - The government should support studies to develop quantum-resistant cryptographic solutions for healthcare applications.
3. **Integration with Emerging Technologies:**
   - Medical data transmission is secured by connecting IoT devices with blockchain-PQC technology in telemedicine along with remote patient monitoring applications.
   - Adding 5G technology to edge computing platforms improves real-time processing security with decentralized EHRs so they become useful for healthcare systems at scale.

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8.3. Summary and Final Thoughts

A revolutionary security method emerged from the combination of blockchain technology with post-quantum cryptography systems to protect EHRs from cybersecurity threats (Kumar et al., 2024). This study investigated the quantum computing obstacles together with conventional cryptography vulnerabilities while showing the future transformational capabilities of quantum-safe decentralized security systems.

8.3.1. Key Contributions of This Study

- The research presented a blockchain-enabled PQC framework which maintains EHR security through decentralized environment where integrity and access control are also protected.
- Traditional cryptographic solutions proved weak against quantum attacks so the research recommended PQC algorithms as defense against current vulnerabilities.
- This research analyzed ethical and legal aspects and regulatory standards that affect decentralized healthcare data management while emphasizing HIPAA and GDPR and other required compliance.
- Security framework implementation and model performance analysis included simulation testing and security assessments as well as traditional security model comparison tests.
- Quantum-secure blockchain protocols and AI-driven security mechanisms as well as interoperability advancements emerged as key future research areas.

8.3.2. Final Thoughts

EHR security advances through unification of blockchain technology with PQC as well as other ongoing advancements in the field. Securing healthcare data management for patients will become possible through continuous research because technological advances are overcoming existing barriers which include computational drawbacks combined with regulatory policy restrictions and difficulties with system adoption (Kumar et al., 2024). Healthcare systems utilizing blockchain-assisted quantum-proof encryption technology transform sensitive health data protection while leading to digital transformation in healthcare secure platforms.

The growth of accessible quantum computing demands proactive development of security frameworks that can protect healthcare infrastructure from post-quantum cyberattacks (Ajakwe et al., 2024). Healthcare providers along with policymakers and technologists must continue their joint efforts to create EHR environments that stay resistant to changing cyber risks while maintaining privacy and security.

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